INCH-POUND

MIL-PRF-6106/10H
16 November 2021
SUPERSEDING
MIL-PRF-6106/10G
24 July 2020

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, NORMALLY OPEN, CONTINUOUS DUTY, (INCLUDING UNSUPPRESSED and SUPPRESSED COILS) 60 AMP, 3PST, ENVIRONMENTALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification sheet and MIL-PRF-6106.

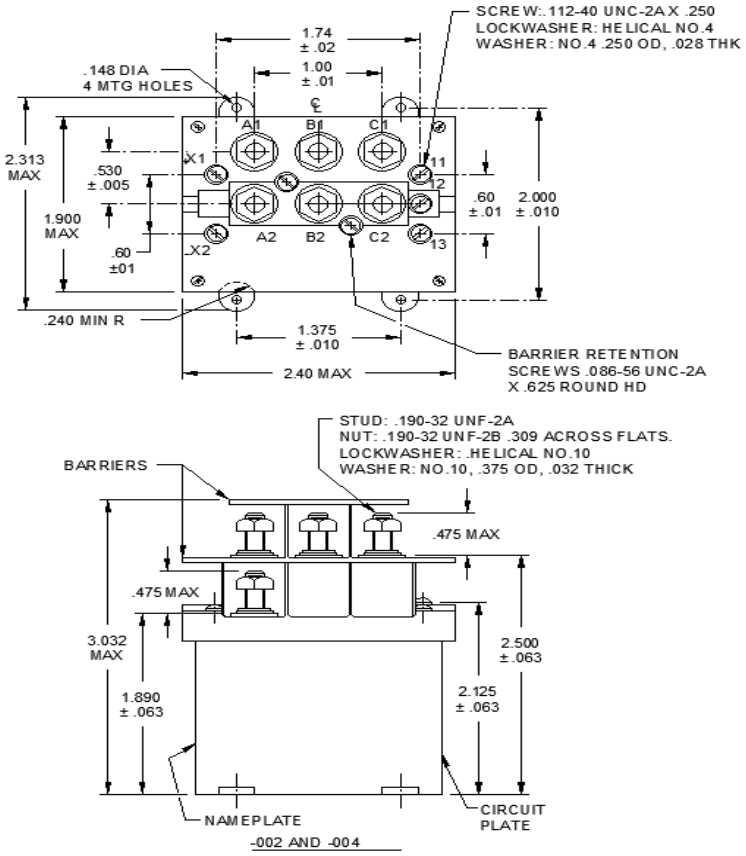


FIGURE 1. Auxilliary Contacts Relay, outline drawing.

AMSC N/A

FSC 5945

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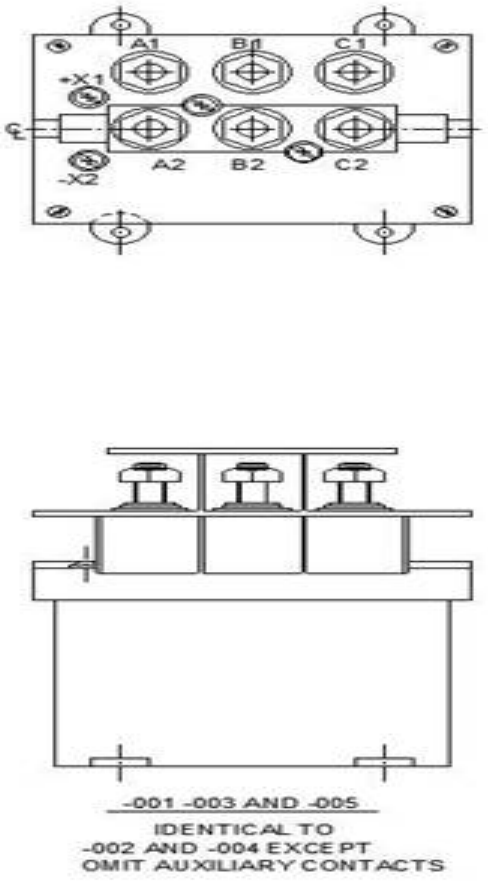


FIGURE 2. Omit Auxiliary Contacts Relay, outline drawing.

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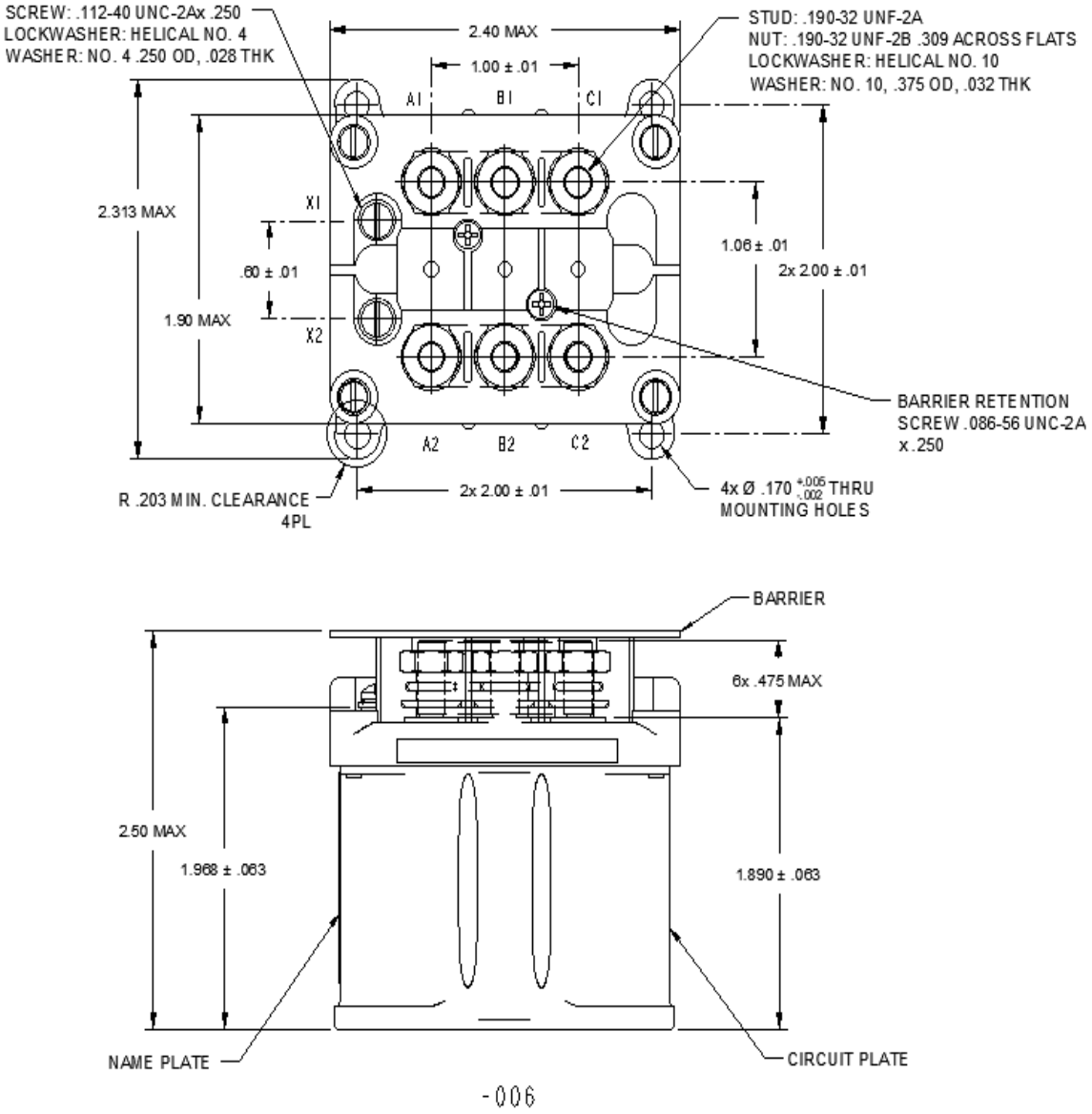
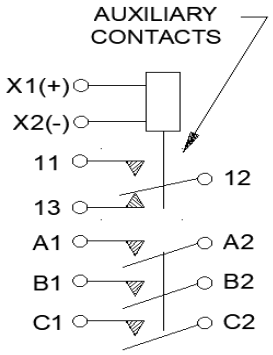
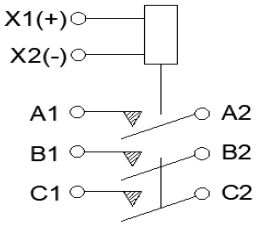


FIGURE 3. Omit Auxiliary Contacts Relay, outline drawing.

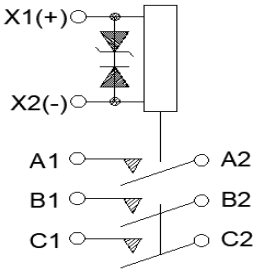
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CIRCUIT
-002 AND -004



CIRCUIT
-001 AND -003
AND -005



CIRCUIT
-005
SEE NOTES 7 AND 8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Weights include cover, barriers, and terminal hardware.
5. When using relay at rated 60 amps, the use of lugs per Burndy part number YAEV6C-G8 or equal is recommended.
6. Terminal temperature rise under continuous current conditions is 95°C maximum.
7. Semiconductor devices shall be JANTX or equivalent. Diodes shall have a peak inverse voltage of 800 V dc minimum.
8. -005 relays shall not operate or be damaged by reversed polarity.

FIGURE 4. Relay, outline drawing.

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TABLE I. Dash number and operating characteristics.

Dash no.	Type	Coil data												Time milliseconds-maximum					
		Coil	Nominal			Maximum			Max pickup voltage 1/			Hold voltage 1/	Drop out voltage 1/	Operate 2/	Release 3/	Bounce			
			Volts	Freq	Res	Volts	Ampere		Normal 1/	High Temp Test	Cont Current Test					Main		Auxiliary	
							Inrush	Steady state								NO	NC	NO	NC
-001	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	-	-	-
-002	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	-	2	2
-003	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	-	-	-
-004	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	-	2	2
-005 4/	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	-	-	-	-
-006	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	-	-	-

1/ Over the temperature range 2/ With nominal coil voltage 3/ From nominal coil voltage
4/ -005 is same as -001 except for transient suppressed coil.

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles $x10^3$	28 V dc 1/				115 V ac, 1 phase 1/				115/200 V ac, 3 phase 1/			
		Main		Auxiliary		Main		Auxiliary		Main		Auxiliary	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	100	-	-	3	3	60	-	3	-	60	-	-	-
Inductive	100	-	-	1.5	1.5	60	-	1.5	-	60	-	-	-
Motor	50	-	-	-	-	40	-	-	-	40	-	-	-
Lamp		-	-	.5	.5		-	.5	-		-	-	-
Transfer load 2/		-	-				-		-		-	-	-
Mechanical life reduced current	200	-	-	.75	.75	15	-	.75	-	15	-	-	-
Mixed loads 3/	50	-	-	Per spec		4	-	-	-	4	-	-	-

1/ Absence of value indicates relay is not rated for that application.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating.
3/ Mixed loads to be conducted at +71°C. Number of mixed loads cycles following continuous current test - 5000.

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REQUIREMENTS

Weight: 001 and 002 and -005: 10 ounces maximum.
003 and 004 and 006: 11 ounces maximum.

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet. 1/

Shock g-level: .5 sine, 25 g's.

Duration: 6 to 9 ms.

Maximum duration contact opening: 1 ms.

Vibration-sinusoidal: See table III.

TABLE III. Vibration-sinusoidal.

Dash No.	Vibration level					
	5-10 Hz	10-55 Hz	55-400 Hz	400-800 Hz	800 - 2000 Hz	
					-55°C and 25°C	+71°C
-001	.08 DA	.06 DA	10 g's	10 g's	10 g's	7 g's
-002			10 g's	8 g's	8 g's	7 g's
-003			10 g's	10 g's	10 g's	7 g's
-004			10 g's	8 g's	8 g's	7 g's
-005			10 g's	10 g's	10 g's	7 g's
-006			10 g's	10 g's	10 g's	7 g's

Vibration (random): Not applicable.

Acceleration: 15 g's.

Seal:

Applicable specification: MIL-STD-202-112, test condition C, procedure IV.

Leak rate: $6x10^{-4}$ std cc/s.

Strength of terminals and mounting studs:

Torque:

Main terminals: 24 inch-pounds maximum, 18 inch-pounds recommended installation torque.

Auxiliary terminals: 4.4 inch-pounds maximum, 3.3 inch-pounds recommended installation torque.

Insulation resistance, initial. 200 megohms minimum.

After life or environmental tests. 100 megohms minimum.

1/ Application range can be extended to 80,000 feet by suitably encapsulating the terminals.

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Dielectric withstanding voltage (sea level): 2 to 5 seconds.

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250	1,500	1,000	1,125
Auxiliary contacts	1,250	1,250	1,000	1,000
Across open power contacts	1,250	1,250	625	625
All other points	1,800	1,800	1,350	1,350

Dielectric withstanding voltage (altitude): 1 minute.

	28 V dc	115 V ac
Coil to case	500	500
Auxiliary contacts	500	500
All other points	700	700

Maximum contact drop, initial: .150 volt

After life test: .200 volt.

Overload current: 320 amperes (ac). 1/

Rupture current: 400 amperes (ac).

Coil suppression (transient voltage, back emf): 45 V dc maximum.

Duty rating: Continuous.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461.

Part number. M6106/10- (dash number from table I).

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-202-112 MIL-STD-461

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